

Janvisree Puligundla

Data Analyst

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PROFESSIONAL SUMMARY

Highly skilled **Data Enthusiast** with over 3 years of experience in leveraging **data analysis, data science, machine learning, and data visualization** tools to solve complex business problems. Proven ability to design scalable pipelines, develop predictive models, and create actionable insights from large datasets. Adept in **statistical analysis, data preprocessing**, and advanced **SQL** scripting to drive business growth and operational efficiency. Seeking to bring expertise in **Python, machine learning frameworks**, and **data visualization platforms** to a forward-thinking organization.

TECHNICAL SKILLS

Programming Languages: Python, SQL (advanced), SAS, C++, Java

Machine Learning: Regression, Classification, Clustering, Ensemble Learning, Neural Networks, Deep Learning, NLP, Generative AI

Data Manipulation: NumPy, Pandas, PySpark

Visualization Tools: Tableau, Power BI, Quicksight, Matplotlib, Seaborn, Plotly, Excel(Pivot tables, VLookup index match, HLookup)

Big Data & Cloud: AWS (S3, Redshift, Glue, SageMaker, IAM), Hadoop, Hive, Spark, Databricks, Snowflake

Databases: MySQL, PostgreSQL, MongoDB, Cassandra, SQLite

DevOps 7 CI/CD: Docker, Kubernetes, Jenkins, Git

Build Tools and IDE's: VS Code, Visual Studio, Eclipse

API Documentation Tools: Postman

Other Skills: Data structures and algorithms, Linux, A/B Testing, Statistical Hypothesis Testing, Time Series Analysis, Feature Engineering, Data Wrangling, Predictive Modeling, Dimensionality Reduction, Text Analytics

EDUCATION

Master of Science in Computer Science, **GPA: 3.8**

Aug 2022 –

Dec 2023 **University of North Carolina at Charlotte**

Coursework: Algorithms and Data Structures, Visual analytics, Data visualization, Big data, Database Systems, Knowledge discovery in databases

PROFESSIONAL EXPERIENCE

Jr Data Scientist | McKinsey & Company, Alpharetta, Georgia, US

June 2024 –

Present

- Designed **scalable machine learning models** for optimizing search functionality, leveraging **TensorFlow, Keras**, and **Scikit-learn**, resulting in a 20% increase in search efficiency.
- Automated **ETL pipelines** to preprocess and transform 100+ GB of data from diverse sources, improving data readiness and analysis speed by 30%.
- Conducted **advanced A/B testing** and **statistical hypothesis testing** using **python (SciPy, statsmodels)** enabling data-backed insights that enhanced customer engagement metrics by 25%.
- Engineered a comprehensive framework for **hyperparameter optimization**, improving model performance by 25%.
- Developed real-time dashboards in **Tableau** and **Python (Plotly, Seaborn)** to visualize KPIs, reducing reporting time by 30%.
- Collaborated with engineering teams to deploy machine learning models using **Docker, Kubernetes**, and **AWS SageMaker** enhancing scalability and deployment reliability.

Tech Stack: Python, TensorFlow, Scikit-learn, PySpark, AWS (S3, Glue, SageMaker, IAM), Tableau, Docker, Kubernetes

Data analyst | JP Morgan Chase, Charlotte, NC, USA

Aug 2023 –

May 2024

- Built **ETL pipelines** using **Python, AWS Redshift, Glue**, and **S3**, ensuring seamless integration of loan repayment data across systems.
- Developed **advanced SQL queries** and **views** for real-time reporting in **Tableau**, enabling the creation of **dynamic dashboards** that reduced stakeholder interpretation time by 20%.
- Applied **machine learning algorithms (Random Forest, Logistic Regression)** for customer behavior prediction, achieving a **15% improvement in accuracy** and increasing quarterly revenue by \$50,000.

- Extracted and transformed data from **SQL databases** and **AWS S3, Redshift**, enabling advanced analysis for loan repayment strategies and improved collection rates by 12%.
- Conducted **predictive modeling** for loan default risks using **Python (Scikit-learn)** and **Tableau**, achieving a precision score of 92%.

Tech Stack: Python, SQL, Tableau, AWS (Redshift, Glue,S3), Scikit-learn, Pandas, NumPy, Plotly, seaborn, Matplotlib, Pyspark

Data analyst/Firmware Engineer | Hewlett Packard, Bangalore, India

Jan 2021 – Jul 2022

- Conducted detailed analysis of **sales trends, inventory data**, and **regional performance** using Python libraries (**Pandas, NumPy, Matplotlib**), enabling a 15% increase in annual revenue.
- Developed and implemented a **predictive maintenance system** that improved operational efficiency by 25% and reduced downtime by 20%.
- Built advanced models using **Random Forests** to predict CI/CD pipeline behavior, improving release reliability by 20%.
- Designed predictive models to **refine pricing strategies**, boosting revenue growth by 12%.
- Created reports on inventory levels and customer satisfaction scores, leveraging **Tableau, python** for visualization and SQL for querying databases.
- Designed a predictive maintenance system with **Random Forest models** and **AWS SageMaker**, improving on-time maintenance execution by 25% and reducing downtime by 20%.

Tech Stack: Python, SQL, Tableau, Scikit-learn, AWS SageMaker, Pandas, Matplotlib

PERSONAL PROJECTS

Chronic Kidney disease prediction

- Developed a predictive model to assist in diagnosing chronic kidney disease using supervised learning algorithms, including Logistic Regression, Random Forest, and SVM.
- Conducted extensive data preprocessing with Pandas to handle missing values, normalize data, and encode categorical variables.
- Optimized the model's performance using feature engineering, hyperparameter tuning with GridSearchCV, and cross-validation to achieve an accuracy improvement of 25%.
- Combined top-performing models into an ensemble using a single neuron called perceptron further boosting prediction efficiency by 20%.
- Visualized insights into the dataset and feature importance using Matplotlib and Seaborn

Environment: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

IPL Score and Performance Analysis Dashboard

- Built an interactive Tableau dashboard to analyze IPL match data, including team performance, player statistics, and season trends.
- Aggregated and cleaned data using SQL and Python, ensuring accuracy and consistency for effective visualization.
- Designed advanced filters and interactive charts, allowing users to explore data by teams, players, venues, and match types.
- Highlighted metrics like points tables, player contributions, and dismissal types, enabling stakeholders to identify key insights and patterns.

Environment: Tableau, Python, Pandas, SQL

Customer Churn prediction

- Designed a machine learning model to predict customer churn for subscription-based services using Logistic Regression and Random Forest.
- Performed exploratory data analysis (EDA) to uncover trends in customer behaviors and their correlation with churn.
- Enhanced prediction accuracy by implementing feature selection, data normalization, and hyperparameter tuning.
- Created a user-friendly Tableau dashboard to visualize customer segments and risk scores, providing actionable insights for churn reduction strategies.

Environment: Python, Pandas, NumPy, Scikit-learn, Seaborn, Tableau